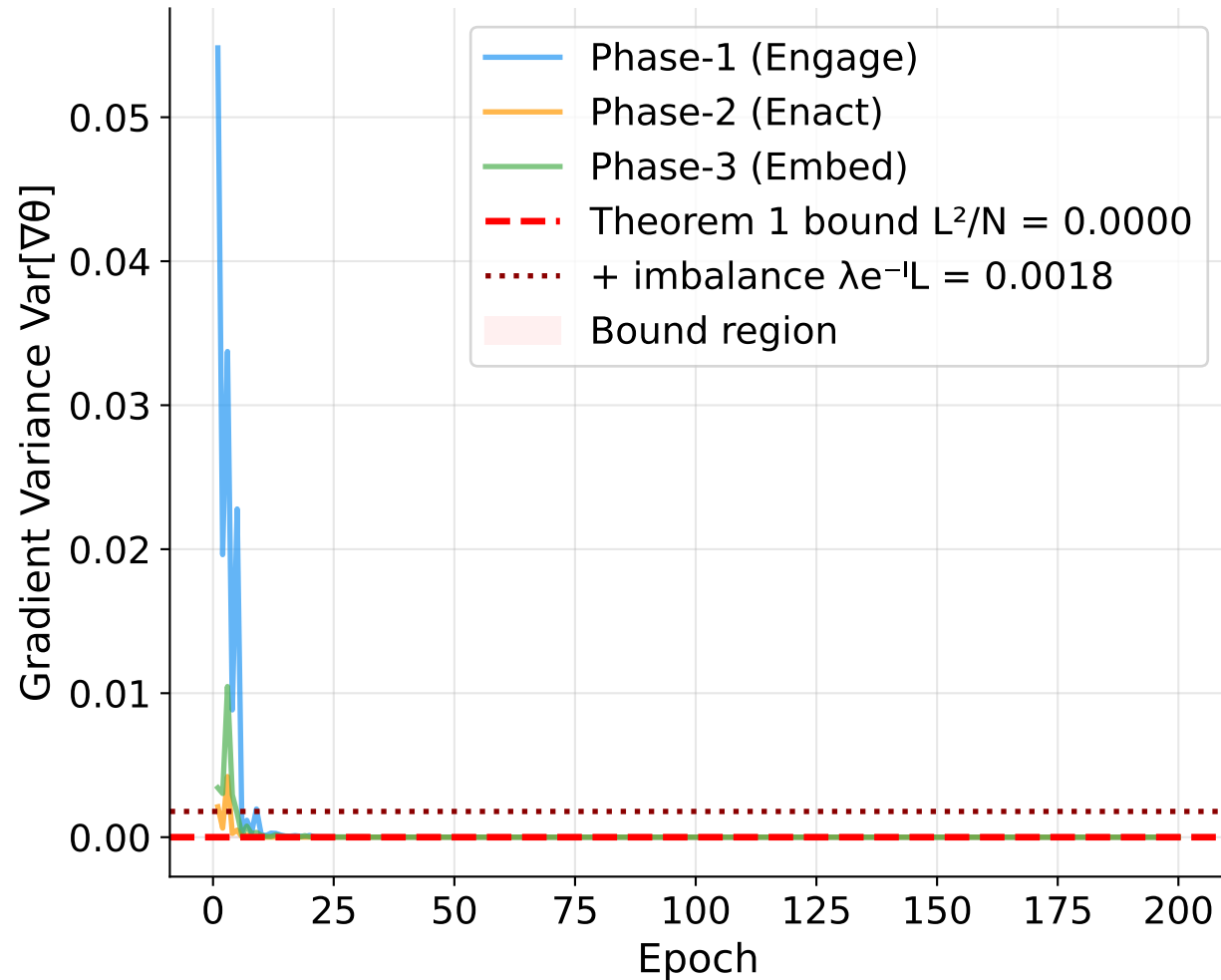
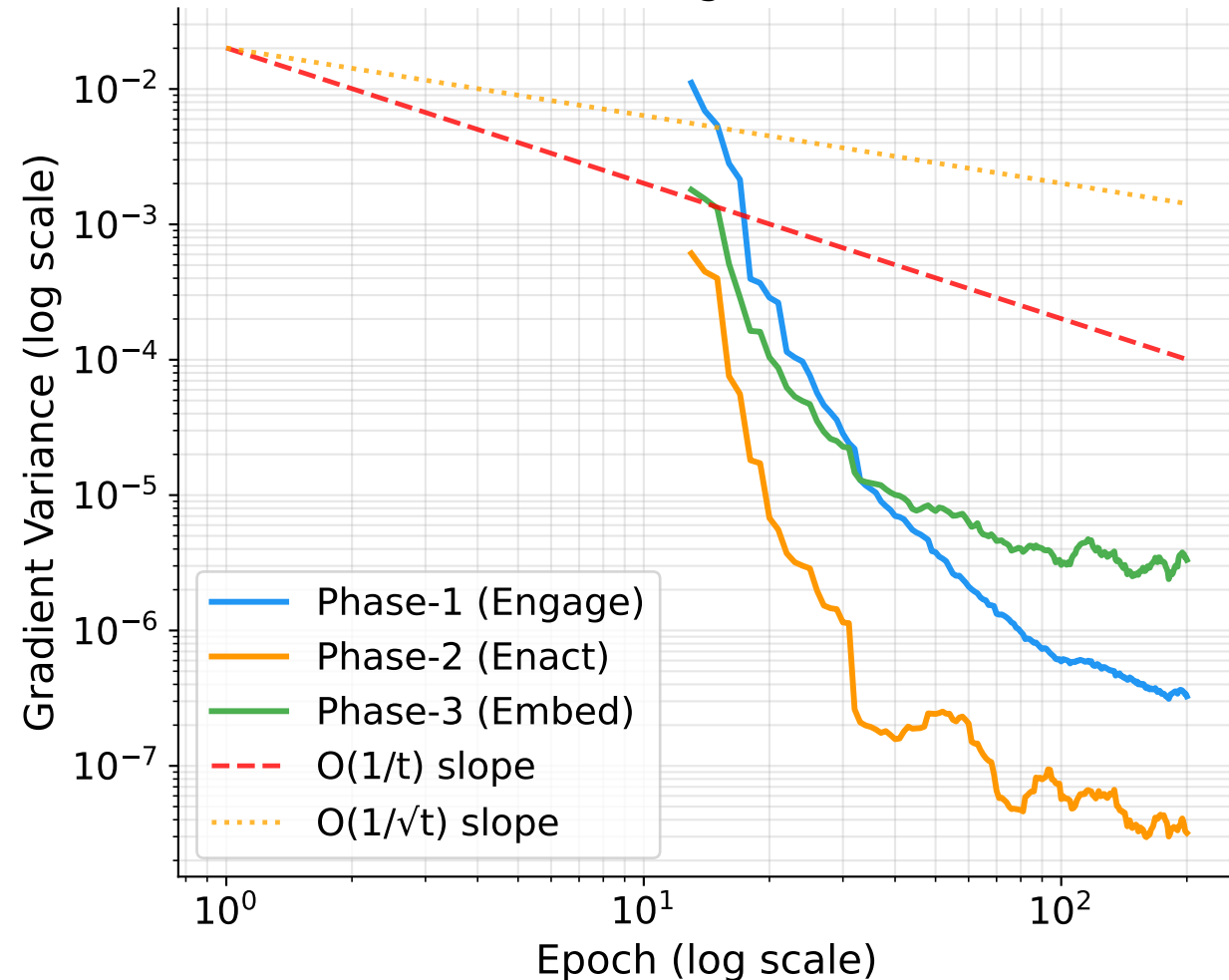


Empirical Variance vs Theorem 1 Bound  
 $\text{Var}[\nabla] \leq L^2/N$



Variance Decay Rate — Log-Log  
Confirms  $O(1/t)$  convergence rate (Theorem 1)



Imbalance Penalty Validation  
Engage phase excess  $\leq \lambda e^{-L}$  (Theorem 1 term 3)

